Reproduction of Merchant, Arnold, Baumgart, Mukherjee, Pein & Pinker (ISPSD 1991)

"*Realization of High Breakdown Voltage (>700 V) in Thin SOI Devices*"
Proc. ISPSD 1991, pp. 31-35

Model vs paper

Oxide breakdown, t_s=0.1um t_ox=2um	model=1210 V	paper=1200 V (Fig.4 dashed line)
BV, t_s=0.1um t_ox=2um, W->inf	model=863 V	paper=~865 V (Fig.5 plateau)
BV, t_s=0.1um t_ox=2um, W=70um	model=849 V	paper=750 V (Fig.3, L=70um device)
Design W for BV=750V (Fig.3)	model=49.4 um	paper=< L=70 um (self-consistent)

Generated by build_report.py -- see derivation.md for the full derivation.

Theory I -- the optimum-field SOI RESURF design

A lateral P+-N diode in a thin SOI film sits over a buried oxide (BOX) on a grounded N+ substrate (shorted to P+). x is lateral (0 at the P+/N junction), y vertical downward. t_s, t_ox = SOI / oxide thickness, L = P+-to-N+ spacing, W = lateral depletion width at breakdown (W<L).

The design goal

Instead of starting from a doping profile and solving for the field, the paper starts from the field distribution that maximizes breakdown voltage for a given W -- a laterally UNIFORM E_x, independent of y -- and works backward through Poisson's equation to the doping N(x) that produces it. The result (eqs 1-9 of the paper):

$$E_y(x,y) = V \, x \, y \,/\, (t \, W \, t_s) \qquad (15)$$

$$E_x(y) = -(V/W)\left(1 - y^2/(2 \, t_s \, t) \right) \qquad (16)$$

$$t = t_s/2 + (K_s/K_{ox}) \, t_{ox}, \quad K_s/K_{ox} = 11.7/3.9 = 3 \qquad (8)$$

t is the effective series thickness of half-depleted silicon plus 3x the oxide (the Si/oxide displacement-continuity jump). E_x peaks at y=0; E_y peaks at the corner (x=W, y=t_s). Integrating the resulting Q(x) shows the required lateral doping N(x) is LINEAR in x -- the SOI analogue of the 'variation of lateral doping' (VLD) concept.

Closed-form potential (used for Fig. 3)

$$\phi(x,y) = (Vx/W)\left(1 - y^2/(2 \, t \, t_s) \right) \quad 0 \le x \le W, \; 0 \le y \le t_s$$

and phi ~ xy in the charge-free oxide beneath the depletion region (E_y vertically uniform, E_x laterally uniform there).

Theory II -- avalanche breakdown

Ionization only occurs in silicon, via the paper's own Chynoweth fit:
alpha(E) = a exp(-b/E), a=7.03e5 cm^-1, b=1.468e6 V/cm
(valid 1.75e5 <= E <= 6.4e5 V/cm; a different b than Nakagawa 1990).

Two dominant ionization paths

Path 1 -- generated at y=0 where |E_x| is largest (and, by design,
constant across 0<x<W): identical to a 1-D p-i-n diode of width W.

$$I1 = a\, W \exp(-b\, W / V) \qquad (17)$$

Path 2 -- generated at the corner (x=W, y=t_s) where |E_y| is largest:
vertical leg through the Si depth (closed form via exponential
integral Ei) plus a horizontal leg back along y=t_s.

$$I2' = a\, t_s\, g\, [Ei(-g) + \exp(-g)/g], \quad g = b\, t / V \qquad (20,21)$$

$$I2'' = a\, W \exp(-b\, W / (V\, (1 - t_s/2t))) \qquad (22)$$

Breakdown: the lowest V solving (1-I1)(1-I2'-I2'')=0, i.e. the lowest
V at which max(I1, I2'+I2'') first reaches 1.

Why thinner SOI can raise the breakdown voltage

As W->infinity, I1 and I2'' vanish, leaving only I2'(V,t_s,t_ox)=1 --
a BV(t_s,t_ox) relation independent of W (Fig.4). A shorter silicon
path (smaller t_s) needs a higher critical field to reach ionization
integral 1 -- exactly as in a one-sided step junction. Below
t_s~1um this effect overtakes the 'shorter path' effect, so BV rises
again as t_s falls -- the 'U' shape of Fig.4 -- until the buried
oxide's own dielectric strength (600 V/um) caps the achievable BV.

Fig. 3 -- equipotential contours (t_s=0.1um, t_ox=2um, L=70um)

Paper

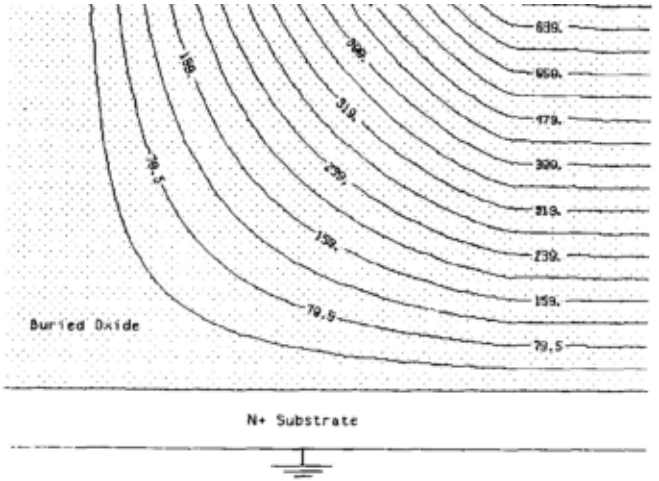
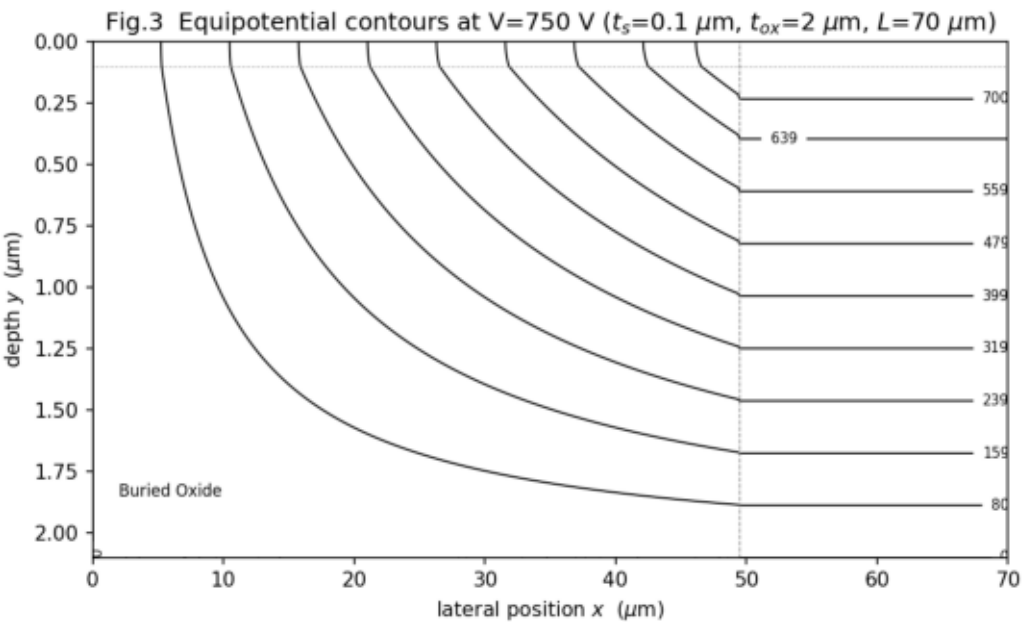


FIG. 3: Two-dimensional off-state numerical simulation of the SOI diode of Fig. 2 with $t_s = 0.1$ μm, $t_{ox} = 2$ μm, and $L = 70$ μm (plot not to scale). Equipotential contours are shown for reverse bias voltage of 750 V, which, as determined by numerical evaluation of the ionization integrals, is the breakdown voltage.

Model



Model: closed-form phi(x,y) plotted directly (no numerical Poisson solve needed); design W=49.4 um found by inverting BV(W)=750V.

Fig. 4 -- breakdown voltage vs SOI thickness (W->infinity limit)

Paper

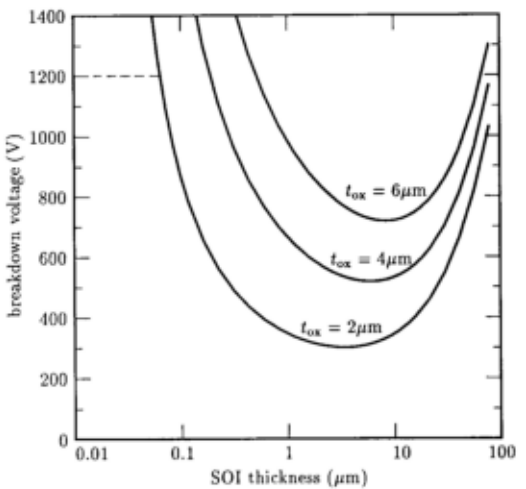
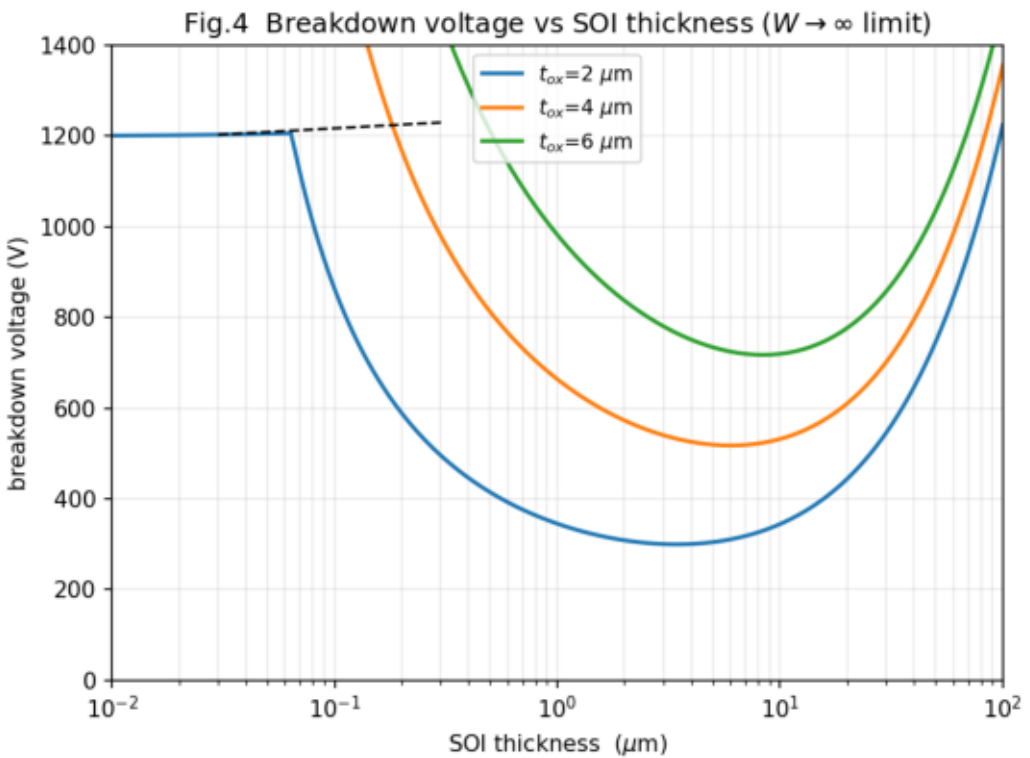


FIG. 4: Theoretical breakdown voltage versus SOI thickness, for three values of buried oxide thickness t_{ox}. This graph is valid in the limit where the lateral depletion width is large. The dashed line corresponds to dielectric breakdown of the 2 μm-thick buried oxide.

Model



Model reproduces the characteristic 'U' shape and the oxide-breakdown cap (1210 V modeled vs 1200 V paper) closely; absolute BV level is ~15% high, likely from graph-reading/rounding tolerance in a,b.

Fig. 5 -- breakdown voltage vs lateral depletion width W

Paper

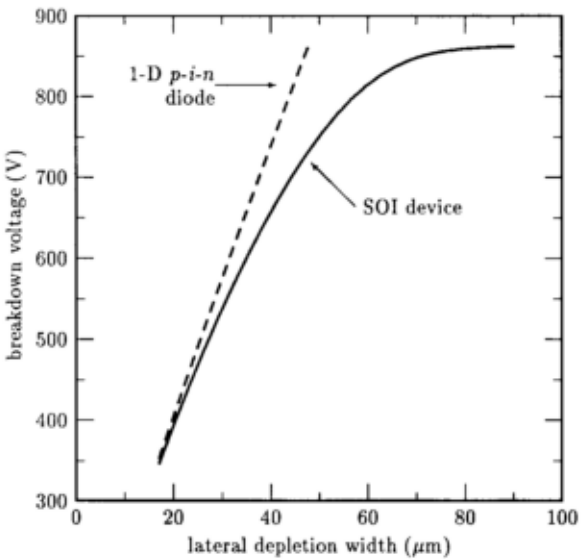
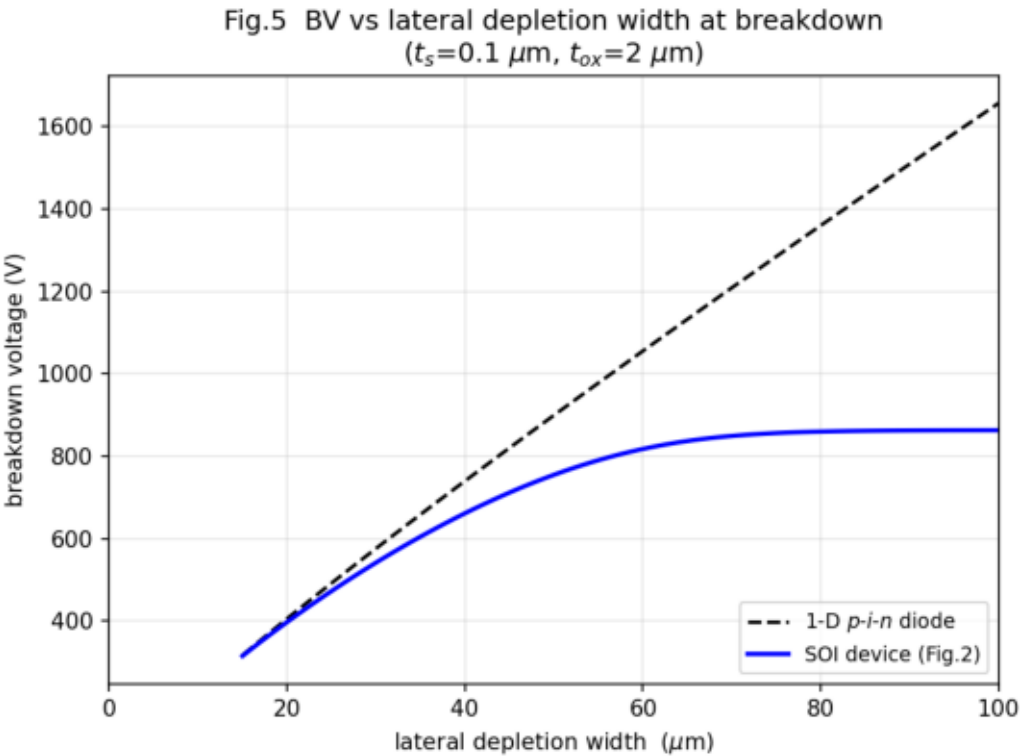


FIG. 5: Theoretical breakdown voltage versus lateral depletion width at breakdown. The solid curve represents the SOI device of Fig. 2, where $t_s = 0.1$ μm and $t_{ox} = 2$ μm. The dashed curve represents the one-dimensional p-i-n diode.

FIG. 6
diode.
5 μA
referri

Model



Excellent match: both the SOI-device saturation (~863-870 V) and the linear 1-D p-i-n reference track the paper closely over the full range.

Extension -- 4H-SiC and GaN films

Nothing in the derivation is silicon-specific except the single-carrier Chynoweth ionization fit alpha(E)=a exp(-b/E) and the film's relative permittivity eps_r (the buried oxide stays SiO2). Swapping in literature Chynoweth fits for 4H-SiC and GaN re-evaluates the SAME closed-form model for a wide-bandgap-on-insulator device.

Material parameters used

Si	eps_r=11.7	a=7.03e5 cm^-1	b=1.468e6 V/cm	(paper's own fit)
4H-SiC	eps_r=9.7	a=3.25e6 cm^-1	b=1.71e7 V/cm	(Raghunathan & Baliga, APL 72, 3503, 1998)
GaN	eps_r=8.9	a=6.4e5 cm^-1	b=1.454e7 V/cm	(Cao et al., APL 112, 262103, 2018)

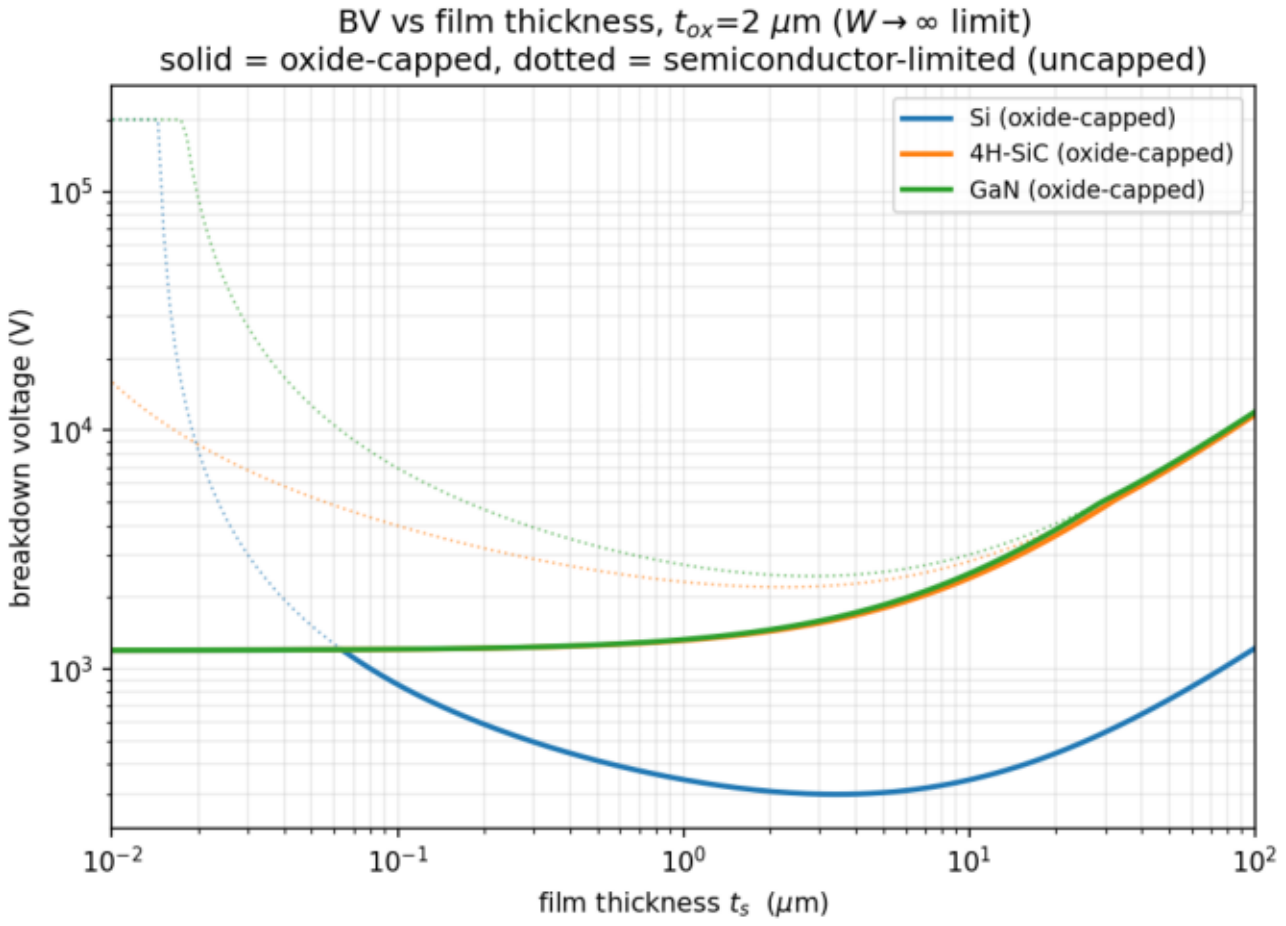
Results at t_s=0.1um, t_ox=2um, W->inf

material	semi-limit BV	oxide cap	actual BV	crossover t_s
Si	863 V	1210 V	863 V	0.064 um
4H-SiC	3990 V	1212 V	1212 V	32.808 um
GaN	6947 V	1213 V	1213 V	28.793 um

Key finding

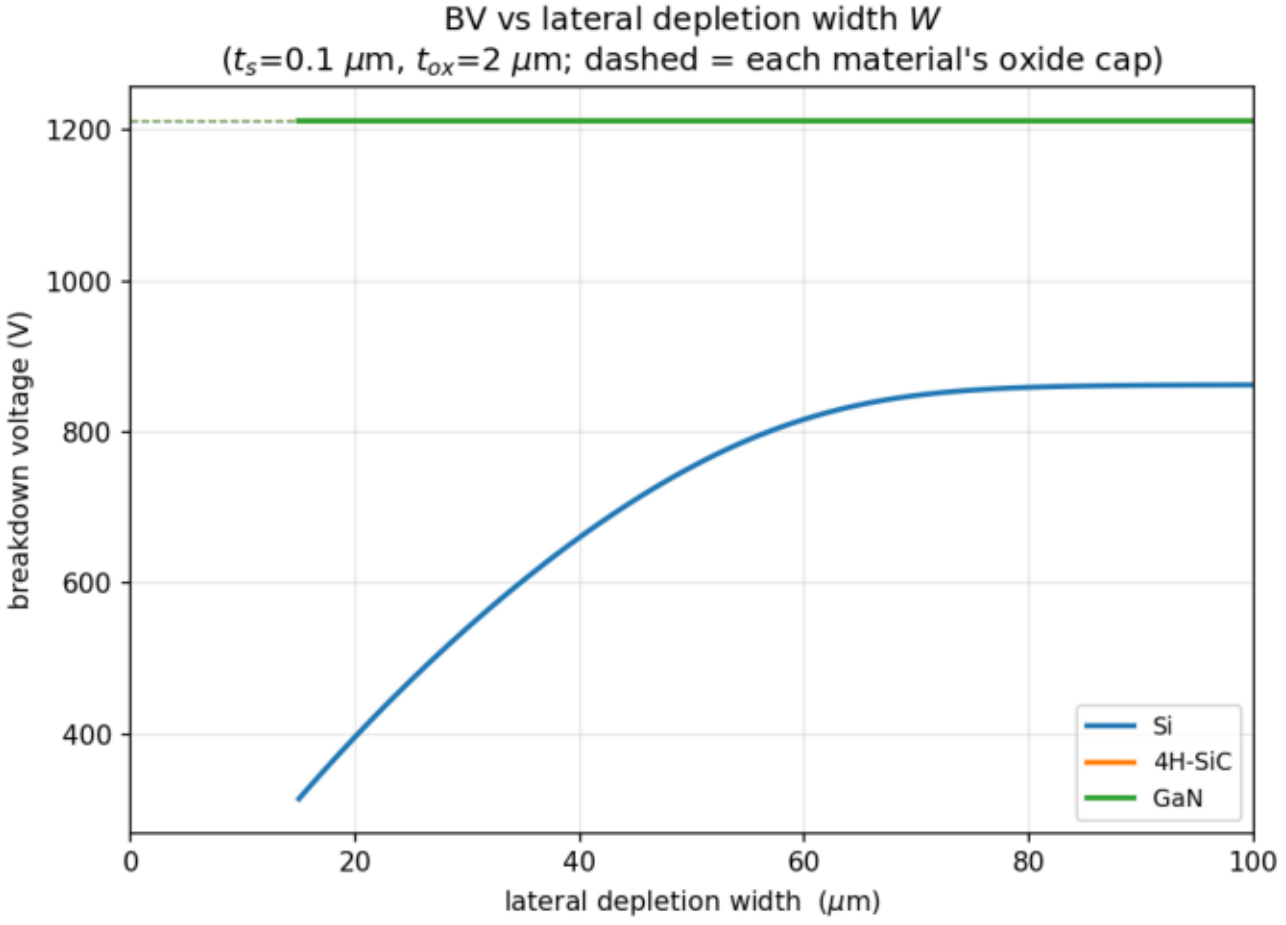
At the paper's own thin-film design point, 4H-SiC's/GaN's much higher critical field is entirely wasted: the buried SiO2 breaks down first, capping all three materials at ~1210-1213 V. The oxide/film crossover thickness is EXACTLY independent of t_ox (the common factor t cancels out of bv=V_ox -- a closed-form fact of the model, not a coincidence; see derivation.md sec.6 and crossover_film_thickness() in model.py). For 4H-SiC/GaN this crossover sits at ~29-33 um -- two orders of magnitude above the sub-micron regime that makes the paper's Si result interesting. Thinning a wide-bandgap film the way this paper thins silicon does NOT raise its breakdown voltage here; only a thicker or higher-strength buried dielectric would let its advantage show through.

BV vs film thickness -- Si / 4H-SiC / GaN (tox=2um)



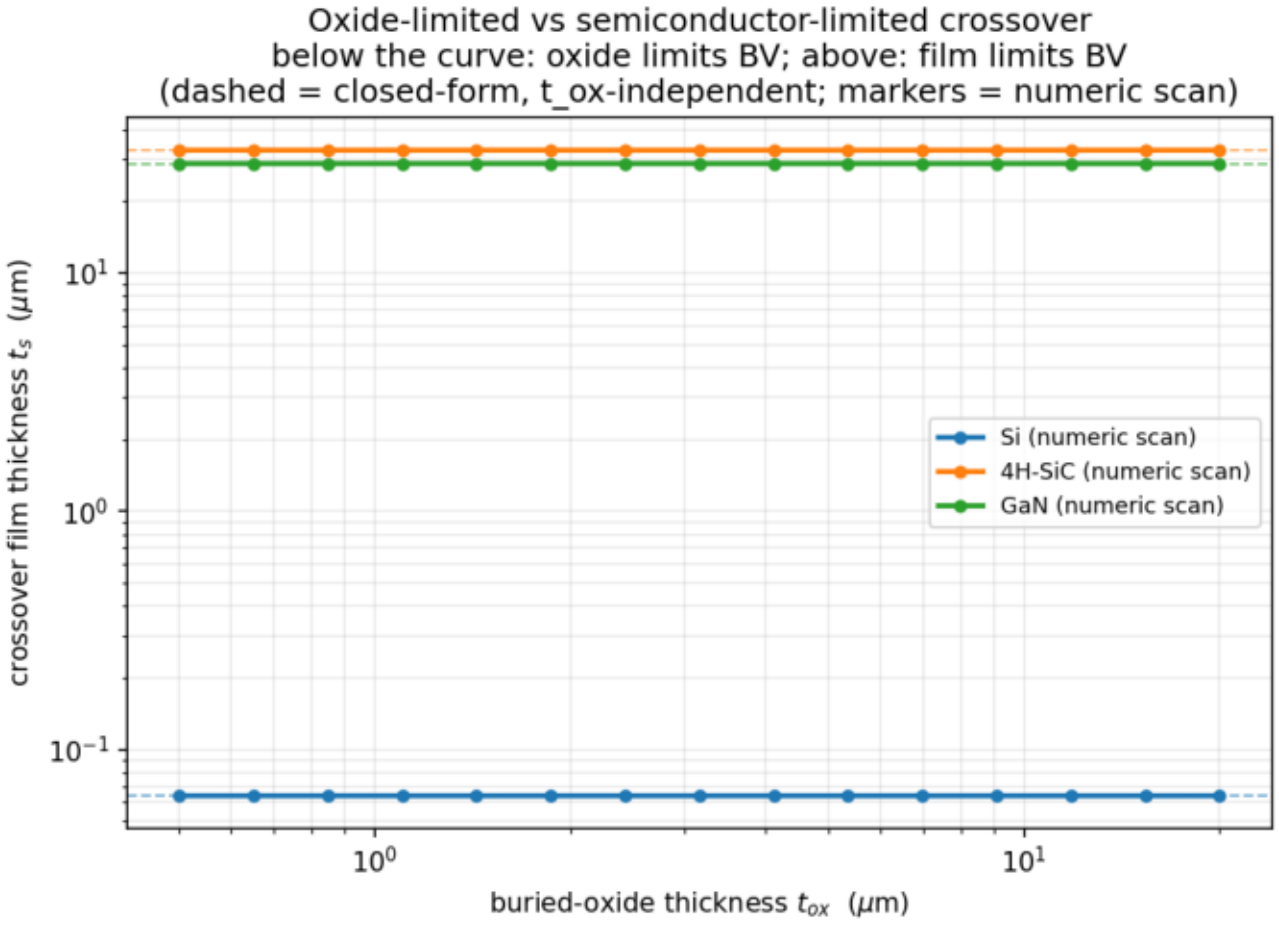
Solid = oxide-capped (actual), dotted = semiconductor-limited (uncapped). Model-only extension -- no corresponding paper figure.

BV vs lateral depletion width -- Si / 4H-SiC / GaN



t_s=0.1um, t_ox=2um. Dashed = each material's oxide cap. Model-only extension -- no corresponding paper figure.

Oxide-limited vs semiconductor-limited crossover



Crossover film thickness t_s is exactly independent of t_ox for all three materials (closed form matches brute-force numerical scan).

The plateau problem -- why Si's BV(t_s) minimum is broad

V_BR(t_s,t_ox) factors EXACTLY into a t_ox-independent critical-field term and a purely geometric term:

$$V_{BR}(t_s,t_{ox}) = E_c(t_s) * t_{eff}(t_s,t_{ox}), \quad E_c(t_s) = b/g^*(t_s)$$

where g*(t_s) solves a*t_s*g*[Ei(-g)+exp(-g)/g]=1 -- an equation in t_s and the material's ionization prefactor a ALONE (model.g_star, model.critical_field). Neither t_ox nor eps_r enters E_c(t_s) at all.

Minimum condition

Taking logarithmic derivatives, dV_BR/dt_s=0 where

$$-d\ln(E_c)/d\ln(t_s) = (t_s/2) / (t_s/2 + r*t_{ox}), \quad r = \varepsilon_r/\varepsilon_{ox}$$

LHS depends on material and t_s only (not t_ox); RHS depends on material's r, t_s and t_ox (not on a or b). Confirmed against direct numerical minimization of BV(t_s): the crossing locates the minimum to within 0.01 um for all 9 (material, t_ox) combinations tested.

Why Si's plateau is broader

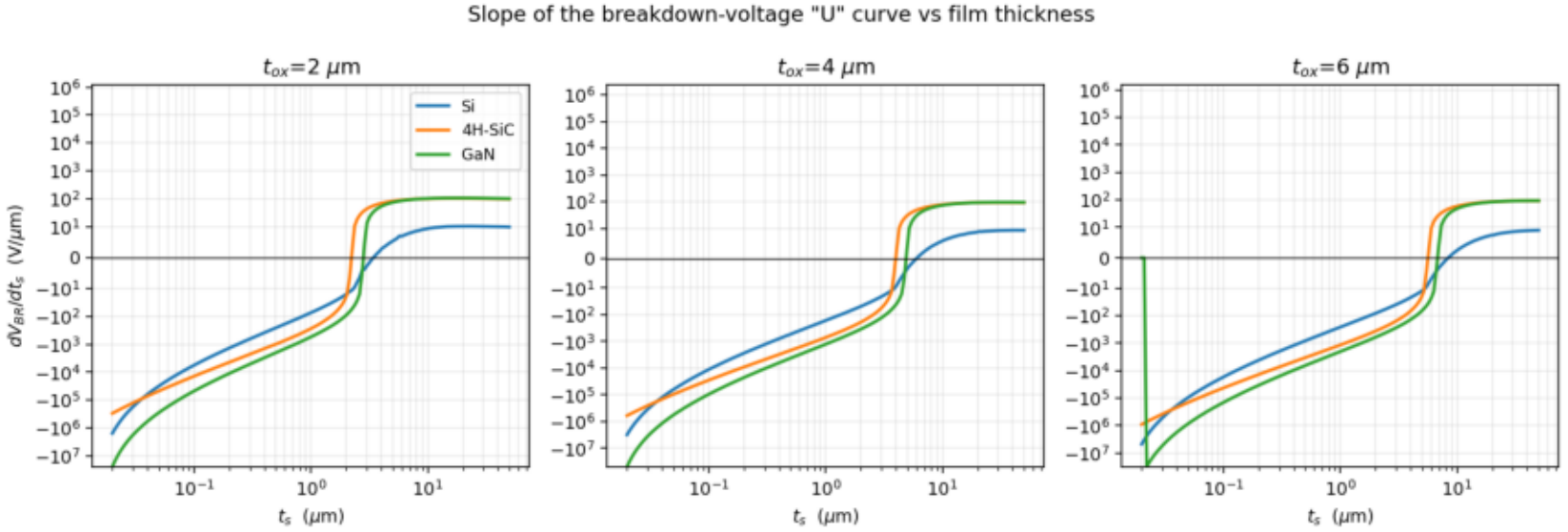
Si's Chynoweth b is ~10x smaller than 4H-SiC's/GaN's, so its critical field's log-slope (the LHS) varies far more gently with t_s. A gently-varying LHS stays nearly matched to the (slowly-varying, 0-1 bounded) RHS over a WIDE t_s range -- a broad plateau. A steeply-varying LHS (4H-SiC, GaN) crosses sharply at one point -- a narrow 'V'.

Quantified: plateau width (|dV_BR/dt_s|/BV_min < 1%/um)

t_ox	Si	4H-SiC	GaN
2 um	1.19 um	0.58 um	0.73 um
4 um	4.47 um	2.27 um	2.78 um
6 um	13.03 um	5.24 um	6.34 um

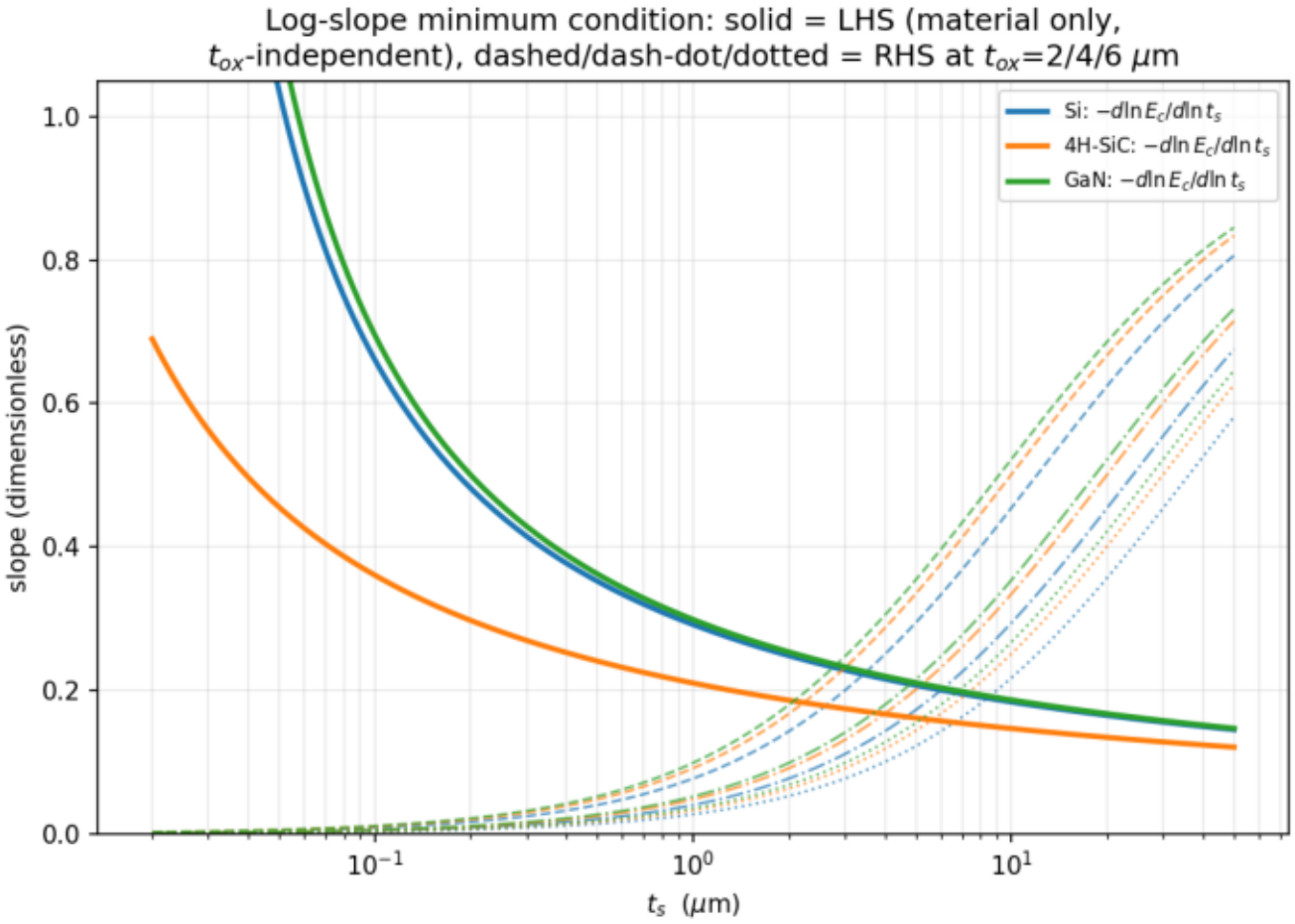
Si's plateau is consistently ~2x wider than 4H-SiC's/GaN's at the same t_ox. See derivation.md sec.7 for the full derivation.

dV_BR/dt_s vs t_s -- Si / 4H-SiC / GaN, per t_ox



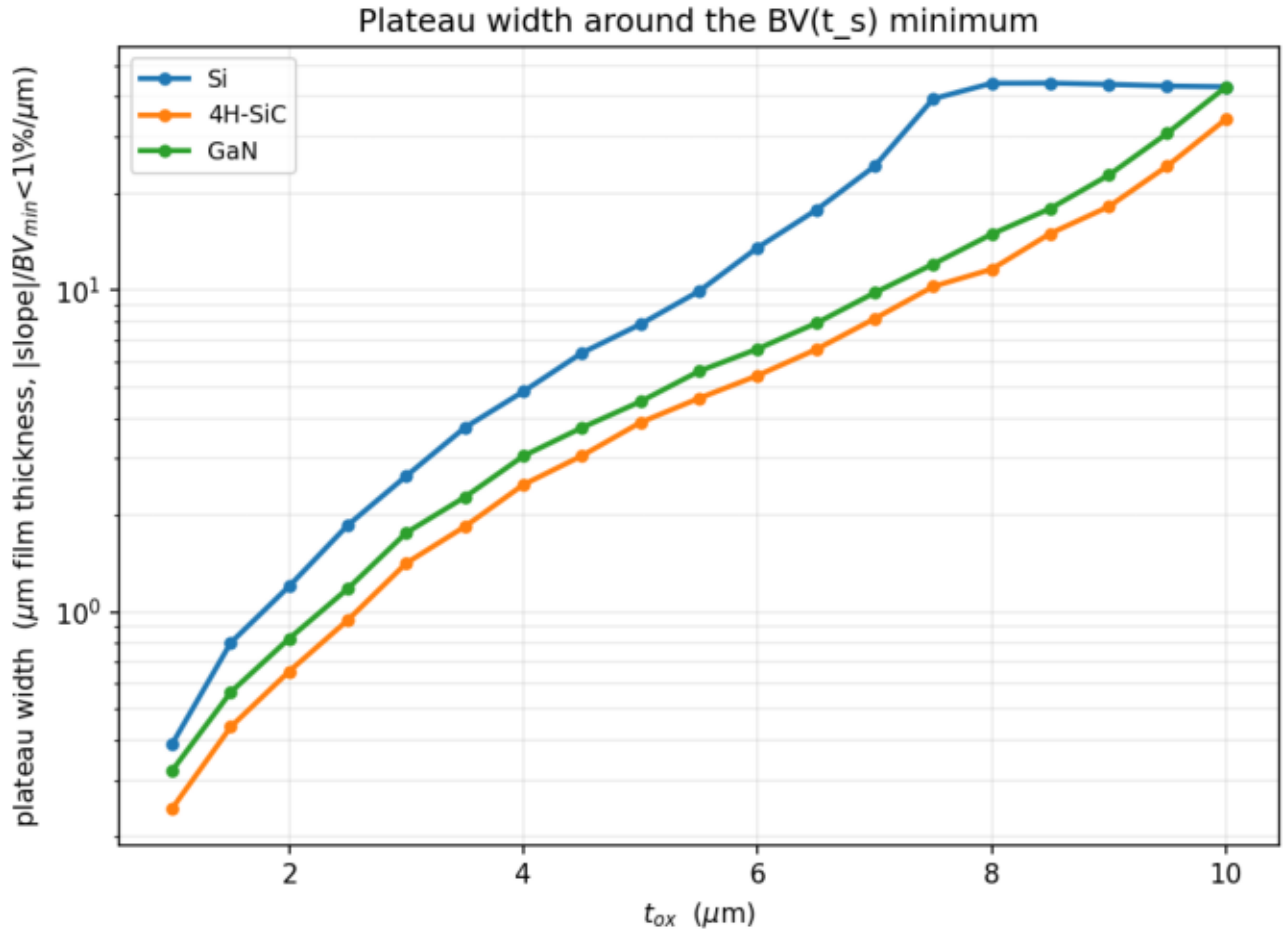
Raw derivative of the breakdown-voltage 'U' curve. The wider the flat region around the zero-crossing, the broader the plateau.

Log-slope minimum condition (LHS/RHS of the boxed equation)



Solid = LHS (material-only, t_ox-independent); dashed/dash-dot/dotted = RHS at t_ox=2/4/6um. Crossings match BV(t_s)'s numerical minima.

Plateau width vs t_ox, per material



Si's plateau is consistently about 2x wider than 4H-SiC's or GaN's at the same t_ox, for the reason derived above.

Notes and limitations

Fig. 6 (experimental I-V trace) -- out of scope

Fig. 6 is a photographed oscilloscope trace of a fabricated device; it is experimental data, not a simulation target. See figs_paper/fig6_paper_crop.png for the original.

Known quantitative gap

The absolute breakdown-voltage level in Fig. 4 runs ~15% higher than the paper's text ('700 V may be achieved' at t_s=0.1um, t_ox=2um); however Fig. 5's W->100um plateau (~865 V) and the oxide-breakdown cap (1210 vs 1200 V) both match closely, so the paper's '700 V' text is most likely a rounded reference to the >700V experimental headline rather than a precise Fig.4 graph reading -- see derivation.md sec.4 and fig4_bv_vs_ts.py's printed comparison.

Assumptions

- Ionization coefficients a, b are the paper's own stated fit (eq.10), not the Van Overstraeten-de Man values used in later papers.
- Fig.3's device doping is assumed to exactly follow the derived linear N(x) out to a single design width W; beyond W the SOI is treated as quasi-neutral at the applied bias.
- Only the two paper-identified dominant ionization paths are evaluated (eq. 13's product formally has infinitely many paths).